# Machine Learning-Driven Cultivar Identification: A Performance Study of Multinomial Logistic Regression and Linear SVM

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***Abstract*—This work compares the performance of Multinomial Logistic Regression (MLR) and Linear Support Vector Machines (SVM) for the classification of wine cultivars according to their chemical composition. Using a common pipeline and 10-fold cross-validation, 30 different model setups were evaluated. The findings suggest that although both approaches are highly effective, the probabilistic nature of MLR is slightly more stable (99.29% accuracy) than the geometric approach (98.62%).**

***Keywords*—Multiclass classification, Multinomial Logistic Regression, Support Vector Machines (SVM), Chemical Profiling, Hyperparameter Optimization, Regularization, Wine Data Analysis, StandardScaler.***

## I. Introduction

Multiclass classification is a common supervised learning problem, in which a model is trained to classify an input into one of several possible classes. This is in contrast to binary classification, where only two classes are involved, and the model must establish complex boundaries to isolate several clusters in a multidimensional space. This is usually done either by probabilistic approaches, where the probability of class membership is estimated, or by geometric approaches, where boundaries are set up to separate data points.

In analytical chemistry, chemical profiling plays an important role in quality control and verification of origin [1], [2]. Chemical profiling refers to the quantitative determination of the chemical components that are unique to a substance (such as alcohol, phenols and flavonoids), thereby establishing a "fingerprint" for the substance [2]. Using machine learning, these profiles can be used to accurately identify the botanical and geographical sources of products, like wine.

The main goal of this study is to conduct a comparative study of a probabilistic decision-making model (Multinomial Logistic Regression) and a geometric decision-making model (Linear Support Vector Machine). Through the optimization of these models using a set of 13 chemical features of three wine cultivars, this research seeks to determine which mathematical approach offers a more stable and reliable generalisation for chemical profiling.

## II. Methodology

The computational code for this study was written in the Python programming language with the implementation and evaluation of the models in the Scikit-Learn library [1]. The approach is based on a typical machine learning process to achieve reliable chemical profiling results.

### A. Dataset Splitting

The data set of 178 chemical samples was divided into a training and testing set using an 80/20 stratified split. This allocated 142 samples for training (used to fit the model and tune hyperparameters) and 36 samples for "hold-out" testing (used to assess the final model's performance). Stratification was used to maintain the same distribution of the three wine varieties in both subsets.

### B. Data Scaling

The 13 chemical features being studied, including Alcohol (measured in percentage) and Proline (measured in mg/L), vary greatly in their magnitudes. To avoid features with larger absolute values having a greater impact on the cost functions of the models, StandardScaler was used. This normalises the features by removing the mean and scaling to unit variance by:

$$z = \frac{x - \mu}{\sigma}$$

Where *mu* is the mean of the feature and *sigma* is the standard deviation. This step is especially important for the Linear SVM as it uses distance metrics to determine the geometric margin.

### C. Pipeline Architecture

To ensure the integrity of the experiments, and prevent data leakage, the scaling and classification models were wrapped within a Scikit-Learn Pipeline. This ensures that the StandardScaler estimates the mean and variance solely from the training folds in the cross-validation process, and not from the test folds or the final test set.

## III. Experimental Design

Here we provide details of the theoretical and mathematical basis of the two models compared. The hyperparameters of both models were selected by performing a 10-fold cross-validation to ensure the selected parameters were well suited to the chemical data.

### A. Multinomial Logistic Regression (MLR)

The MLR model is a probabilistic classification model that generalises the binary logistic regression to multiple classes using the Softmax function. This converts the linear combination of the 13 chemical features into a probability distribution over the three wine cultivars, given by:

$$P(y = j|\mathbf{x}) = \frac{e^{\mathbf{x}^\top \mathbf{w}_j}}{\sum_{k=1}^{K} e^{\mathbf{x}^\top \mathbf{w}_k}}$$

Where K=3. The main variable in the experimental design for MLR was the Regularization parameter (C). The C parameter in Scikit-Learn is the inverse of the regularization parameter (C=1/λ). A lower C value means a stronger penalty for large coefficients, which results in a simpler model without over-representing certain chemical markers (overfitting) [3]. A larger C value means less penalty, so the model is able to fit complex patterns in the chemical profiles but suffers from high variance.

### B. Linear Support Vector Machine (SVM)

The SVM method is essentially geometric. It aims to find a Maximum Margin Hyperplane that maximises the distance (the "margin") between the support vectors (the chemical samples nearest the decision boundary) [4].

The SVM optimization was controlled by the Cost (C) parameter, which controls the "soft margin". A larger C value will ensure all training points are classified correctly by making the margin narrower (overfitting), resulting in a less robust model for new data. A small C value permits some training points to be misclassified, but results in a larger, more stable margin. To guarantee that the geometric coordinates of the hyperplane were globally converged, the maximum number of iterations (max_iter) was set to 10,000, as typical chemical data may require more iterations to find the optimal boundary.

## IV. Comparative Results

### A. Hyperparameter Optimization and Tuning Trends

The models were tuned by an extensive grid search using 10-fold cross-validation. Both models were found to be extremely sensitive to the regularization constant C, with all models achieving maximum accuracy at C=0.1. As shown in Table I, the Multinomial Logistic Regression (MLR) model attained a higher mean accuracy (*99.29%*) with the smallest variance (*0.0214*), implying a highly robust decision boundary.

TABLE I. Comprehensive MLR Tuning Results

| Tested C Value | Comprehensive MLR Tuning Results | | |
|---|---|---|---|
| | ***Solver*** | ***Mean CV Accuracy*** | ***Std. Deviation*** |
| 0.001 | lbfgs / saga | 0.7538 | 0.0898 |
| 0.01 | lbfgs / saga | 0.9714 | 0.0349 |
| **0.1** | **lbfgs** | **0.9929** | **0.0214** |
| 1.0 | lbfgs | 0.9862 | 0.0276 |
| 10.0 | lbfgs | 0.9790 | 0.0320 |
| 100.0+ | lbfgs | 0.9719 | 0.0344 |

TABLE II. Comprehensive SVM Tuning Results

| Tested C Value | Comprehensive SVM Tuning Results | | |
|---|---|---|---|
| | ***Dual Mode*** | ***Mean CV Accuracy*** | ***Std. Deviation*** |
| 0.0001 | False | 0.9300 | 0.0541 |
| 0.001 | False | 0.9504 | 0.0556 |
| 0.01 | False | 0.9790 | 0.0320 |
| 0.1 | False | 0.9862 | 0.0276 |
| 1.0 | False | 0.9719 | 0.0344 |
| 10.0 | False | 0.9647 | 0.0352 |

The empirical results showed that C values less than 0.01 caused underfitting due to over-regularization, and C values greater than 10.0 caused a slight decrease in accuracy and an increase in variance, suggesting overfitting.

### B. Final Test Metrics

After tuning, the models were then applied on the test reservoir of 36 samples. The geometry of the SVM and the probability scores of the MLR both achieved 100% classification accuracy on this data.

TABLE III. Final Performance Metrics on Test Reservoir

| Metric | Final Performance Metrics on Test Reservoir | |
|---|---|---|
| | ***Model A: MLR*** | ***Model B: SVM*** |
| Total Test Samples | 36 | 36 |
| Final Accuracy | 1.0000 | 1.0000 |
| Weighted F1-Score | 1.0000 | 1.0000 |
| Cohen's Kappa Score | 1.0000 | 1.0000 |
| Precision (Weighted Avg) | 1.0000 | 1.0000 |
| Recall (Weighted Avg) | 1.0000 | 1.0000 |

### C. Final Test Metrics

In order to learn how the models make decisions, the feature weights (coefficients) were obtained. Figure 1 shows the comparison of the feature importance for the 13 chemical constituents.

The most important features for wine classification were Proline, Flavonoids and Color Intensity [5]. The feature

weighting indicates that the chemical data is highly ordered, and both the geometric and probabilistic models converged on similar critical features for classification.

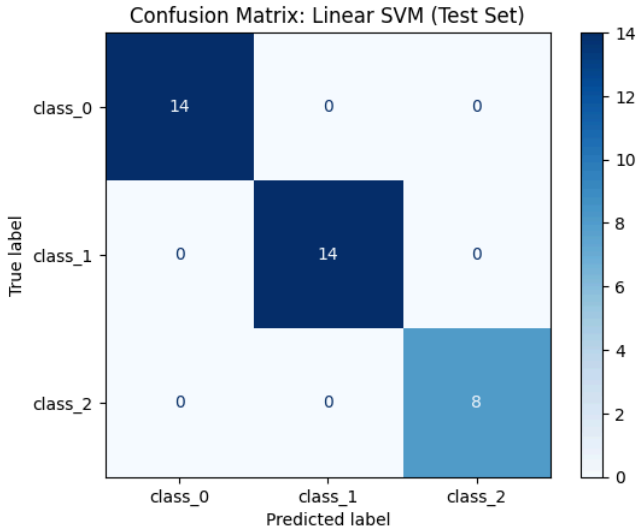


Fig. 1. Confusion Matrix for Linear SVM (Test Set).

As shown in the Confusion Matrix, the SVM model achieved a perfect diagonal, correctly identifying all 14 samples of Class 0, 14 of Class 1, and 8 of Class 2.

## V. Conclusion

The comparative study shows that both the probabilistic and geometric mathematical approaches are extremely effective in discriminating wine cultivars, provided that the data is properly normalized. Although both Multinomial Logistic Regression (MLR) and Linear Support Vector Machines (SVM) models reached the same level of accuracy (100%) on the final testing set, the 10-fold cross validation results show a slight but consistent difference in model stability.

The empirical evidence from the hyperparameter optimization phase leads to the conclusion that the probabilistic model (MLR) is the more stable model for the given chemical data. This is primarily due to two reasons:

- Mean Performance: MLR's mean cross-validation accuracy (*99.29%*) is higher than that of the SVM (*98.62%*).
- Robustness of the Model: The standard deviation of the MLR was much smaller (*0.0214*) than the SVM (*0.0276*), suggesting that the decision boundaries (probability distributions) are less affected by the folds of the data.

Also, the research confirmed the importance of the StandardScaler in the geometric approach. The unit variance prevented the maximum margin of the SVM to be skewed by large features such as Proline. In conclusion, although the geometric separation of SVM is successful, the soft-max probabilistic output of MLR offers better performance and generalization for high-dimensional chemical profiling.

## References


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